BIOT Module 2

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1. Introduction

Alzheimer's disease changes the brain long before anyone notices a forgotten name or a misplaced key. By the time it is diagnosed, the changes are everywhere: in which genes are switched on, in which proteins are made, and in the DNA a person was born with. Each of these leaves its own trace, and each has been measured in large public studies.

Looked at one at a time, each layer gives a long list of genes, and much of each list is noise. A gene that changes in only one layer could be a side effect of the disease, or chance. A gene that stands out in all three is much harder to dismiss.

In Module 1 I looked up a single APOE variant. Here I widen the view from one gene to thousands.

This brings us to the main question of this module: which genes show evidence for Alzheimer's disease in RNA, protein and DNA at the same time, and which of them are worth following as possible drug targets?

2. Disease and data choice, with biological interpretation

I chose Alzheimer's disease (AD) because it is the most common cause of dementia and still has very few treatments, and because it follows on from the APOE variant I looked up in Module 1.

I downloaded all three layers myself with one script (`scripts/get_data.py`). All are free and need no login:

Layer	Source	What it measures	Genes
RNA	Agora (AMP-AD)	AD against healthy brains, 3 donor groups (ROSMAP, Mayo, MSBB), 9 brain regions	20,539
Protein	Agora (AMP-AD)	AD against healthy brains, prefrontal cortex	8,254
Genetics	GWAS Catalog	every published DNA variant linked to AD	3,558

Agora pools three donor groups and nine brain regions, so it is larger than any single study, and the GWAS Catalog collects every published AD genetics result.

One thing shapes everything that follows: the data is unmatched. Each file gives one number per gene, not per person, so I joined the layers by gene name rather than by patient.

What the results mean biologically:

- *Known AD genes come out on top:* APOE (rank 11), the biggest genetic risk factor, rises at both RNA and protein. HLA-DRB1 (2), CLU (31), BIN1 (53), PICALM (83) and SORL1 (220) also appear, so the method finds real AD biology.
- *The top 15 fall into four groups, all known to change in AD:* brain immune cells (HLA-DRA, HLA-DRB1, ITGAX, all up), fat transport (APOC1, APOE, APOB), synapses (RPH3A, NRXN3, SHANK2, all down, which fits synapse loss) and blood vessels (PECAM1, APLNR).
- *Surprise, GMPR (rank 7):* not a known AD gene, yet it rises at both RNA (+1.04) and protein (+0.23), and both results are strong. It is worth a closer look.
- *Missing genes:* TREM2, CD33 and MS4A6A are known AD genes but are not in the list, because the protein study did not measure them.

Key terms:

- *Fold change (lfc):* how much a gene changes in AD against healthy brains. Above 0 is up, below 0 is down.
- *−log10 p:* how strong a genetic link is. Bigger is stronger.
- *Concordant:* RNA and protein move in the same direction.

3. Method

1. Load the three files.
2. For RNA, keep each gene's biggest change across brain regions, skipping the cerebellum, which AD mostly spares.
3. Keep only genes found in all three layers: 1,115 genes (Figure 1).
4. Mark a gene concordant if RNA and protein move the same way: 635 of 1,115 (57%) do.
5. Rank each layer from 0 to 1 (by the size of the change, or the GWAS strength) and average the three ranks with equal weight.
6. Sort by score. Save the top 15 to `targets_ad.csv` and all 1,115 to `targets_ad_full.csv`.

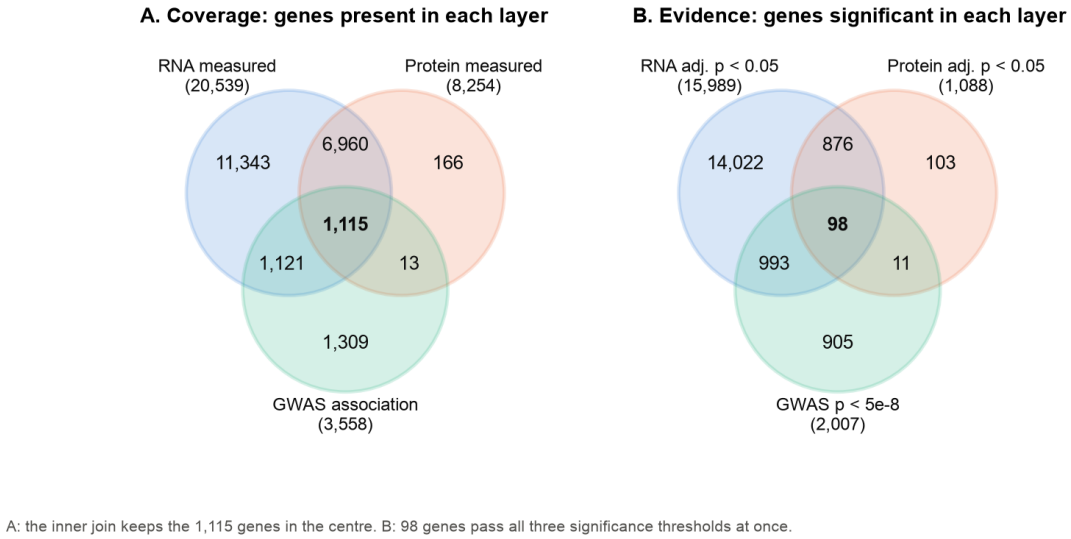


Figure 1: Genes in each layer. A: genes measured; the centre (1,115) is what the integration keeps. B: genes significant in each layer; 98 are significant in all three.

4. Results

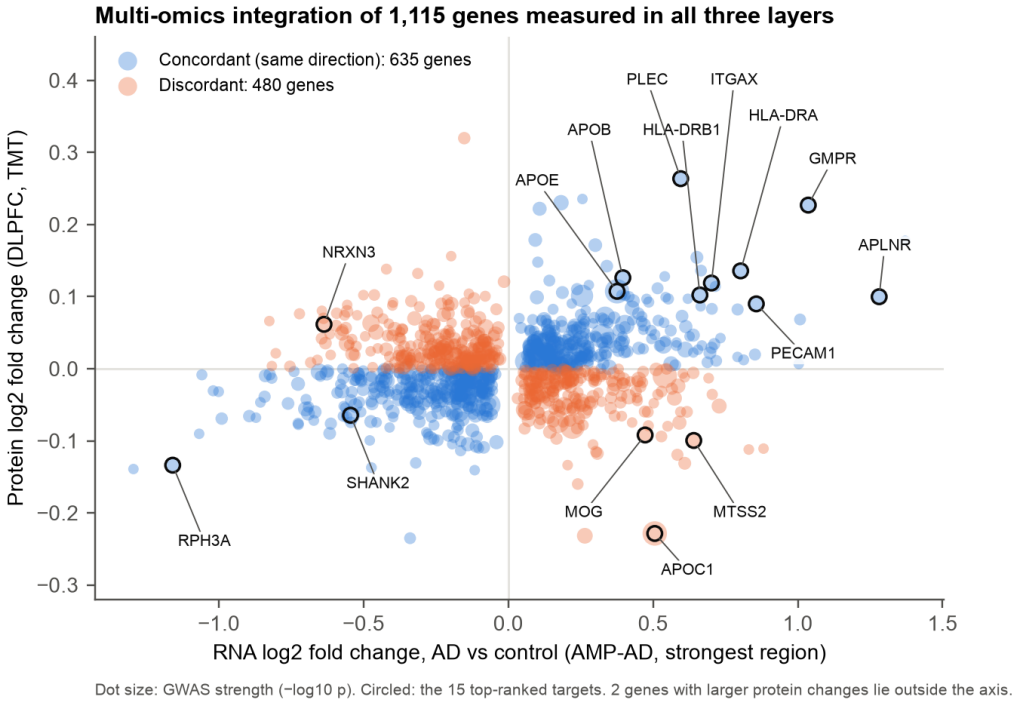


Figure 2: RNA against protein change for the 1,115 genes. Blue moves the same way in both, orange disagrees. Dot size is GWAS strength; the 15 top-ranked genes are circled.

Table 1: Top 15 genes. RNA and Protein are fold changes; Genetics is −log10 p.

Rank	Gene	RNA	Protein	Genetics	Concordant	Score
1	HLA-DRA	0.80	0.14	16.7	Yes	0.957
2	HLA-DRB1	0.66	0.10	22.5	Yes	0.955
3	APOC1	0.51	−0.23	672.7	No	0.949
4	PECAM1	0.86	0.09	16.5	Yes	0.939
5	MTSS2	0.64	−0.10	13.5	No	0.913
6	ITGAX	0.70	0.12	11.7	Yes	0.912
7	GMPR	1.04	0.23	10.1	Yes	0.910
8	RPH3A	−1.16	−0.13	10.0	Yes	0.904
9	NRXN3	−0.64	0.06	22.0	No	0.904
10	APLNR	1.28	0.10	10.5	Yes	0.902
11	APOE	0.38	0.11	496.4	Yes	0.902
12	SHANK2	−0.55	−0.06	30.0	Yes	0.899
13	APOB	0.40	0.13	19.0	Yes	0.897
14	MOG	0.47	−0.09	17.3	No	0.896
15	PLEC	0.60	0.26	10.5	Yes	0.894

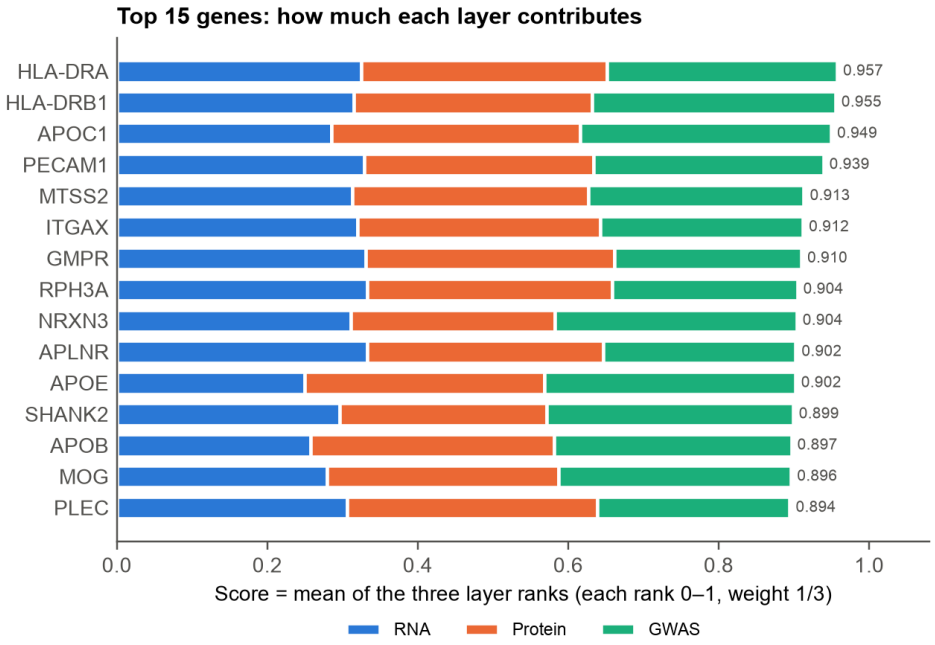


Figure 3: Each top-15 score split into its three layer ranks. Every top gene is high in all three layers, so no single layer carries the list.

5. Discordance and limitations

A gene where RNA and protein disagree, APOC1 (rank 3): its RNA rises (+0.51, parahippocampal gyrus) while its protein falls (−0.23, prefrontal cortex), and both changes are strong. Possible reasons:

1. RNA and protein were measured in different brain regions.
2. APOC1 is a protein that cells send out, so less of it stays in the tissue.
3. The protein may be broken down faster, or trapped in plaques.
4. There may be more immune cells in the tissue, which raises RNA without each cell making more protein.

Its huge genetic score (673) is also probably mostly APOE's signal, because APOC1 sits right next to APOE on chromosome 19.

Not every “No” is real. BIN1 is marked not concordant, but its protein change is tiny and not significant ($p = 0.43$), so that is probably noise.

What I can claim: these genes have several separate kinds of evidence, and the three layers barely track each other (Figure 2), so a high score is not one signal counted three times.

What I cannot claim:

1. Anything per patient. In Part 1 (breast cancer), RNA and protein came from the same tumours, so they could be compared person by person. Here the data is unmatched.
2. What causes what. A genetic link points to a region of DNA, not a proven cause, as APOC1 shows.
3. That the missing genes don’t matter. TREM2 was dropped only because protein wasn’t measured.
4. Exact effect sizes. Taking the biggest RNA change per gene makes RNA effects look larger, and only 15% of genes have a significant protein change.

6. Weighting and conclusion

Weighting: equal weights (1/3 each). Ranking each layer first stops APOE’s huge genetic score from taking over. Protein could get less weight because its changes are small, but I kept equal weights to match the course default.

Conclusion: combining RNA, protein and genetics put known AD genes such as APOE, HLA-DRB1 and CLU near the top, which suggests the method works. The ranked list goes on to Module 3. Genes with strong evidence in all three layers, such as APOE, the HLA genes and GMPR, are the best places to start.

If I did this again, I would:

1. Not require all three layers, so genes like TREM2 are not lost just because protein wasn’t measured.
2. Use RNA and protein from the same brain region.
3. Count a gene as concordant only when both changes are significant.
4. Treat APOE and its neighbours (APOC1, TOMM40) as one genetic signal.